SWS — Director’s 3D Stage

Implementation Specification for the Vibe-Coding App

Purpose: Build a professional, prompt-driven virtual film set and previsualization system inside Stage Work Studio (SWS). The existing SWS video prompt is the primary shot-direction input; the 3D Stage translates it into spatial, camera, performance, continuity and timing information.

Core principle: Do not build a generic 3D editor. Build a director-oriented AI filmmaking previs system.

1. OBJECTIVE

Implement a professional Director’s 3D Stage / Virtual Previsualization System inside SWS.

The stage must use the video prompt already generated by the SWS Video Prompt System as its primary scene-direction input.

The stage must let the director inspect environment, set, characters/mannequins, blocking, camera, lens, camera movement, character movement, facial expression, eye direction, interaction, props, lighting, dialogue, screen direction and spatial relationships.

2. ARCHITECTURAL PRINCIPLE

Do not treat the video prompt as a simple text description. Create a structured intermediate representation called SHOT DIRECTOR DATA.

SHOT DIRECTOR DATA

Shot ID, Scene ID, Sequence ID, Duration; Environment, Set, Time of Day, Weather, Atmosphere; Characters, Positions, Orientation, Scale; Blocking, Movement, Speed, Direction; Expression, Eye Direction, Head Direction, Body Direction, Gesture; Interaction; Dialogue and timing; Camera Position, Height, Rotation, Target; Lens, Sensor, FOV, Focus Distance, Depth of Field, Aperture; Camera Movement, Path, Speed and Duration; Composition, Shot Size, Framing, Screen Position, Foreground/Midground/Background; Lighting; Props; Continuity; Generation Instructions; Original Video Prompt.

3. VIDEO PROMPT PARSER

Parse filmmaking language into structured stage data. Understand camera terms such as dolly, truck, pan, tilt, crane, orbit, tracking and handheld; lens terms such as 24mm, 35mm, 50mm and 85mm; staging, blocking, performance, interaction, environment and dialogue.

Do not merely place generic objects into the stage. Interpret relationships between subjects, camera and environment.

4. 3D MANNEQUIN / CHARACTER SYSTEM

The default mannequin must look beautiful, cinematic, anatomically convincing, proportionally correct, smooth and visually appealing. Do not use an ugly primitive grey stick figure.

Support head, neck, shoulders, arms, hands, fingers, torso, pelvis, legs and feet, with clean neutral cinematic shading.

Support replacement by Character Reference, GLB/GLTF, future high-quality character rigs or custom SWS assets without changing the shot-data architecture.

5. CHARACTER IDENTIFICATION & BLOCKING

Characters retain their SWS identities and names. Support Move, Rotate, Duplicate, Reset, Snap to Ground, Focus, Lock, Hide/Show and Frame Selected.

Persist Position X/Y/Z, Rotation X/Y/Z, Scale, Facing Direction and Movement Path.

6. CHARACTER MOVEMENT

Translate movement descriptions into actual viewport animation wherever possible. Examples: walk toward, turn toward, move background-to-foreground, stop beside another character.

Provide editable straight, curved and spline movement paths with speed, timing, pause, acceleration and deceleration controls.

7. EXPRESSION & EYE DIRECTION

Support expressions including angry, frightened, sad, shocked, calm, determined, suspicious, smiling, crying and neutral, with a structured system ready for future facial rigs.

Track Eye Target, Eye Direction, Head Direction and Body Direction separately. Eye targets may be another character, camera, prop, environment point or custom target.

8. INTERACTION SYSTEM

Represent relationships such as Character A looks at Character B, walks toward Character B, touches Character B, holds a prop or points toward an object. Do not ignore meaningful interaction language.

9. DIALOGUE STAGING

Dialogue is part of shot choreography. Store Speaker, Listener, Dialogue Text, Start Time, End Time, Eye Target, Body Orientation, Expression and Gesture.

Timeline must identify the speaking character and prepare for future lip sync, facial animation, voice generation and AI video generation.

10. CAMERA SYSTEM

Implement a real cinematic perspective camera, not a fake viewport movement. Store Position, Rotation, Target, Focal Length, Sensor, FOV, Focus Distance, Depth of Field and Aperture.

11. CINEMATIC LENS SYSTEM

Support real focal-length presets: 14mm, 18mm, 24mm, 28mm, 35mm, 40mm, 50mm, 65mm, 85mm, 100mm, 135mm and 200mm, plus custom focal lengths.

Changing focal length must actually change perspective and composition.

12. SHOT SIZE & COMPOSITION

Support Extreme Wide, Wide, Full, Medium Full, Medium, Medium Close-Up, Close-Up, Extreme Close-Up, Over-the-Shoulder, Two Shot and POV.

Shot size should be derived from actual camera-to-subject relationships where possible.

13. CAMERA MOVEMENT

Support Static, Pan, Tilt, Dolly In/Out, Truck Left/Right, Crane Up/Down, Orbit, Tracking, Follow, Push In, Pull Out and Handheld simulation.

Each movement must store Start, End, Duration, Speed, Easing and Target. Display the camera path and allow editing.

14. ENVIRONMENT SYSTEM

Create a basic spatial 3D environment rather than a flat background. Support ground, walls, buildings, trees, roads, furniture, rocks, vehicles, architectural elements and atmosphere.

Use procedural primitives where dedicated assets are unavailable, with future GLB/GLTF environment support.

15. ENVIRONMENT FROM PROMPT

Interpret environment language such as ancient village street at sunset and construct a reasonable previs environment communicating space, scale, composition and atmosphere. Final-quality assets are not required at this stage.

16. LIGHTING

Support key, fill, rim, practical, sun and ambient/environment lighting. Interpret moonlight, warm sunset, harsh sunlight, candlelight, blue night light, backlight and silhouette.

17. TIMELINE

Provide a shot timeline with Play, Pause, Stop and Scrub. Show character movement, camera movement, expression changes, eye targets, dialogue, interactions and lighting changes.

18. DIRECTOR VIEW

Create a clean director-oriented viewport. Show the camera frame, characters, composition and essential camera/shot information without clutter.

19. CAMERA FRAME & GUIDES

Support 16:9, 2.39:1, 2:1, 4:3, 1:1 and 9:16. Provide optional Rule of Thirds, Center Cross, Horizon, Safe Area, Action Safe, Title Safe and Center Marker.

20. PROMPT ↔ STAGE SYNCHRONIZATION

Maintain a two-way relationship: Video Prompt ↔ Shot Director Data ↔ 3D Stage.

If the director changes the stage, update Shot Director Data and offer an explicit Update Video Prompt action. Do not silently rewrite the prompt.

21. PROMPT ACCURACY

Priority order: (1) explicit prompt instruction, (2) Shot Director Data, (3) existing scene/character data, (4) previous-shot continuity, (5) reasonable inference.

If information is inferred, mark it INFERRED. If ambiguous, mark it Needs Direction rather than inventing a filmmaking decision.

22. CONTINUITY

Understand previous and next shots. Preserve character position, screen direction, set, props, camera relationship, lighting and orientation unless the new prompt explicitly changes them.

Provide Match Previous Shot and previous/current shot comparison with optional ghost/overlay mode.

23. GENERATION & COMFYUI CONNECTION

The Director’s Stage is part of the existing SWS generation pipeline, not an isolated feature. Its structured shot data must be available to image generation, video prompt generation and ComfyUI workflow generation.

Do not break existing Writer Console, Scene Matrix, Shot Matrix, Video Prompt System, Character System, World System, Project Parser, ComfyUI integration, project storage, naming system or Simple/Advanced switcher.

24. PROJECT FOLDER EXPORT & NAMING

When exporting, use the exact same base name as the ComfyUI workflow. Do not use random serial numbers or UUID-based user-facing filenames.

EXPORT EXAMPLE

Use the existing SWS project-folder architecture. Example:

25. EXPORT FILES

Director Stage: <workflow-name>_DirectorStage.json

Stage Preview: <workflow-name>_StagePreview.png

Video Prompt: <workflow-name>_VideoPrompt.txt

Optional Camera Data: <workflow-name>_Camera.json

Optional Shot Data: <workflow-name>_ShotData.json

Avoid unnecessary duplicate files when equivalent data already exists in SWS.

26. STAGE PREVIEW

Generate a clean camera-view preview showing camera framing, mannequins, environment, props, lighting and staging. It must look like a useful production reference, not a debug screenshot.

27. UI / SIMPLE MODE / PRO MODE

Use a professional, clean director-oriented interface.

Simple Mode: Characters, Set, Camera, Lens, Movement, Lighting, Dialogue, Timeline, Preview, Export.

Pro Mode: Transform, World Coordinates, Focal Length, Sensor, FOV, Focus, Aperture, Keyframes, Animation Curves, Light Parameters, Camera Path, Technical Metadata and JSON.

28. PERFORMANCE & DATA PERSISTENCE

Keep the viewport responsive with optimized GLB/GLTF, instancing, LOD, texture optimization, efficient animation and GPU-friendly rendering.

Autosave all staging, camera, lens, movement, expression, eye direction, dialogue timing, environment, lighting and props. Provide Undo/Redo and Restore Previous Stage State.

29. DO NOT INVENT FILMMAKING DETAILS

Do not turn 'Rama looks at Kara' into 'Rama walks toward Kara'. Do not turn a slow push-in into an orbit. Do not move a character to a different relationship unless explicitly directed.

The stage is a visual interpretation of the prompt, not an independent director.

30. IMPLEMENTATION ORDER

Phase 1 — 3D viewport; Phase 2 — high-quality mannequin; Phase 3 — character placement/rotation; Phase 4 — environment/props; Phase 5 — camera/lens; Phase 6 — camera movement; Phase 7 — character movement; Phase 8 — eye/head direction; Phase 9 — expression system; Phase 10 — interaction; Phase 11 — dialogue; Phase 12 — timeline; Phase 13 — prompt parser; Phase 14 — two-way synchronization; Phase 15 — continuity; Phase 16 — preview; Phase 17 — project export; Phase 18 — ComfyUI integration.

31. STAGE VALIDATION

Before export, validate: Camera defined; Lens defined; Duration defined; Characters staged; Environment defined; Character movement defined; Camera movement defined; Eye direction defined; Dialogue timing defined; Interaction defined; Prompt synchronized; ComfyUI workflow linked; Project path available.

Warnings should be shown for missing optional information but optional data should not unnecessarily block export.

32. PROFESSIONAL PREVIS TARGET

Target the conceptual standard of professional previs, blocking, cinematography planning and virtual production, while developing an SWS workflow optimized specifically for AI filmmaking rather than copying another application's interface.

33. THE FIVE DIRECTOR QUESTIONS

WHERE? Where are the characters and objects?

WHAT? What are they doing?

HOW? How are they interacting and moving?

FROM WHERE? Where is the camera and what lens is being used?

WHEN? When does each action, movement and dialogue happen?

34. FINAL SWS PRODUCT VISION

Writer Console → Script → Scene → Shot → Video Prompt → Prompt-to-Director Data → Director’s 3D Stage → Director Adjustment → Shot Lock → Image/Video Generation → ComfyUI Workflow → AI Video.

The Director’s Stage must become the visual staging and cinematography authority for every shot in SWS.

REFERENCE DATA FLOW

```
VIDEO PROMPT
      ↓
PROMPT PARSER
      ↓
SHOT DIRECTOR DATA
      ↓
DIRECTOR'S 3D STAGE
      ↓
SHOT PREVIS
      ↓
IMAGE / VIDEO GENERATION
      ↓
COMFYUI WORKFLOW
```

REFERENCE PROJECT EXPORT

```
SWS_Project/
    Scene_05/
        Shot_012/
            Shot_012_ComfyUI.json
            Shot_012_VideoPrompt.txt
            Shot_012_DirectorStage.json
            Shot_012_StagePreview.png
```

REFERENCE SHOT DIRECTOR DATA

```
{
  "shotId": "Shot_012",
  "duration": 8,
  "camera": {
    "focalLength": 35,
    "movement": "slow_push_in"
  },
  "characters": [
    {
      "name": "Rama",
      "eyeTarget": "Kara",
      "bodyDirection": "toward_camera"
    },
    {
      "name": "Kara",
      "bodyDirection": "toward_Rama"
    }
  ],
  "dialogue": [],
  "environment": {},
  "originalVideoPrompt": "..."
}
```

Implementation instruction to the vibe-coding app: First inspect the existing SWS architecture and reuse its data models, storage, naming, project structure and UI conventions. Implement the Director’s 3D Stage incrementally in the stated phase order. Do not break existing functionality. Every feature must remain connected to the shot and generation pipeline.